

INDIANAPOLIS EXECUTIVE, IN, USA

WMO: 720575

Lat: 40.031N

Lon: 86.251W

Elev: 922

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00173

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	5.0	-8.2	3.7	0.3	-2.9	4.9	6.4	25.2	37.1	23.8	31.4	5.2	290	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	90.4	74.1	88.0	73.7	85.6	72.4	77.9	86.2	76.2	84.2	74.6	82.3	9.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	137.3	82.8	73.3	128.3	80.6	72.3	123.6	79.2	42.0	86.1	40.2	84.4	38.6	82.4	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.5	17.4	-9.2	93.9	6.4	3.9	-13.8	96.7	-17.6	99.0	-21.2	101.2	-25.8	104.0
			-9.7	79.9	6.3	2.3	-14.2	81.6	-17.9	82.9	-21.4	84.2	-26.0	85.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	25.6	30.3	41.5	52.1	63.9	71.8	73.3	70.8	65.7	53.7	41.1	32.2	
	DBStd	19.00	12.21	12.20	11.18	9.15	9.01	5.79	5.77	4.90	7.07	9.13	9.83	10.50	
	HDD50	2646	758	561	314	83	8	0	0	0	0	65	298	559	
	HDD65	5659	1220	973	732	396	133	13	5	9	76	367	717	1018	
	CDD50	3354	2	8	50	146	438	653	723	644	473	180	31	6	
	CDD65	893	0	0	2	9	99	217	263	188	98	17	0	0	
	CDH74	7839	0	0	19	85	862	1763	2403	1624	928	153	2	0	
	CDH80	2405	0	0	1	6	234	564	853	441	284	22	0	0	
Wind	WSAvg	7.4	9.4	9.2	9.0	9.3	7.6	6.6	4.8	4.1	4.7	7.0	8.1	8.7	
Precipitation	PrecAvg	40.0	2.6	2.2	3.3	3.9	4.2	3.9	4.4	3.3	3.1	2.8	3.4	3.0	
	PrecMax	53.0	9.6	5.9	8.4	8.2	10.6	8.4	13.9	7.9	8.7	7.9	10.1	7.6	
	PrecMin	29.1	0.3	0.4	0.6	1.0	1.1	0.3	0.6	0.2	0.2	0.2	0.3	0.5	
	PrecStd	6.7	1.8	1.2	1.7	1.7	2.1	1.8	2.9	1.7	2.1	1.5	2.0	1.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	64.8	78.7	81.0	89.9	91.2	96.9	90.4	91.2	82.5	72.3	63.1	
		MCWB	55.6	55.9	64.7	61.9	71.4	72.8	77.2	74.3	72.5	68.5	57.7	57.6	
	2%	DB	53.2	58.8	70.1	76.8	85.6	88.6	90.8	87.9	86.9	78.7	66.2	57.3	
		MCWB	50.9	54.2	59.3	61.2	68.7	72.4	75.8	74.1	71.9	64.7	56.8	55.0	
	5%	DB	48.0	53.9	63.9	72.5	82.3	86.1	88.0	84.5	83.1	73.7	61.1	52.0	
		MCWB	45.2	50.2	56.0	58.7	67.1	71.6	75.3	72.6	70.3	62.7	53.9	49.1	
	10%	DB	42.2	48.0	59.3	68.1	79.3	83.1	84.5	82.0	79.7	69.6	56.6	46.3	
		MCWB	39.4	44.1	52.0	57.0	66.6	70.1	73.9	71.1	67.3	60.9	49.9	43.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	59.0	65.3	66.2	73.7	78.0	80.4	79.4	76.4	72.6	63.0	59.2
			MCDB	59.0	62.5	78.3	75.7	83.5	87.3	89.2	85.6	85.1	79.5	69.8	61.6
2%		WB	50.7	54.5	60.8	63.4	71.5	75.6	78.7	76.6	74.5	68.6	58.8	54.9	
		MCDB	52.4	57.0	67.5	71.2	81.1	84.5	87.4	83.9	83.1	73.7	63.5	57.1	
5%		WB	45.7	50.2	57.4	61.2	69.8	73.8	77.1	74.8	72.4	65.6	54.7	49.0	
		MCDB	47.3	54.3	63.5	68.3	78.7	82.3	85.1	81.8	80.1	70.0	60.7	51.4	
10%		WB	39.4	43.8	53.5	59.1	68.2	72.2	75.2	72.9	70.5	62.2	50.6	43.9	
		MCDB	42.7	47.6	58.6	66.3	76.6	80.2	82.8	79.5	76.9	69.2	55.9	46.1	
Mean Daily Temperature Range	5% DB	MDBR	14.8	15.6	17.1	20.4	19.5	18.5	18.9	20.7	21.9	20.9	17.2	13.2	
		MCDBR	19.0	21.6	23.3	26.2	22.5	21.9	21.7	22.8	25.2	25.6	24.0	18.9	
		MCWBR	16.7	17.9	15.1	14.6	10.2	9.7	9.4	11.6	11.8	14.0	15.9	16.7	
	5% WB	MCDBR	18.4	19.9	20.9	20.3	19.4	18.5	19.1	18.7	20.7	19.5	21.4	18.3	
		MCWBR	17.4	17.7	14.2	13.8	10.1	9.6	9.5	10.7	11.0	12.3	16.0	16.9	
Clear-Sky Solar Irradiance	taub	0.300	0.324	0.345	0.403	0.439	0.435	0.451	0.450	0.413	0.358	0.319	0.299		
	taud	2.424	2.337	2.369	2.213	2.164	2.215	2.174	2.163	2.252	2.403	2.461	2.449		
	Ebn at Noon	275	282	288	276	268	268	262	259	262	267	266	266		
	Edh at Noon	26	32	35	44	47	45	46	46	39	30	25	23		
All-Sky Solar Radiation	RadAvg	551	816	1153	1500	1715	1951	1919	1737	1429	992	658	460		
	RadStd	58	89	96	129	156	121	112	100	115	94	73	51		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	N/A

Nomenclature: See separate page